

Euro Working Group on Transportation Annual Meeting 2025 - EWGT2025

How do costs influence Electric Vertical Take-off and Landing aircraft selection?

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Abstract

In order to find a solution to maintain the market share on the banned ultra-short-haul routes, an airline can consider replacing conventional aircraft with an electric Vertical Take-Off and Landing aircraft (eVTOL). To make a satisfactory choice (among six eVTOLs: Joby S4, Lilium Jet, Terafugia TF-2 Lift+Push, Alice, Jaunt, and Embraer Flexcraft) while dealing with multiple criteria, the Analytic Hierarchy Process (AHP) approach can be applied. The sensitivity of eVTOLs' rankings concerning different pairwise comparisons of the alternatives with respect to costs is analysed, showing the influence of cost changes on aircraft selection. The sensitivity analysis shows that the final rank of six aircraft is sensitive to cost changes. It was conducted 720 experiments, 120 per aircraft having the lowest costs. That means that all possibilities have been exhausted and the set of all feasible solutions has been searched. Moreover, for defined pairwise comparison matrix of criteria selected (the first level of hierarchy), it is revealed that Terafugia TF-2 Lift+Push, can never be chosen as the most desirable solution, Alice and Embraer Flexcraft can never be bottom-ranked, while the other three aircraft can be both the best and the worst solution.

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Peer-review under responsibility of the scientific committee of the Euro Working Group on Transportation Annual Meeting 2025 - EWGT2025.

Keywords: AHP approach; sensitivity analysis; eVTOL selection.

1. Introduction

The International Civil Aviation Organization (ICAO) introduced the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), a market-based measure for emission reduction. To achieve international aviation's

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global ambitious goals in emission reduction, many countries have implemented CORSIA. Aiming to promote the sustainable and green transport sector and combat climate change, the European Union (EU) included the aviation sector in the European Union Trading Scheme (EU-ETS). Some European countries went one step further and prevented ultra-short-haul flights between cities that are linked by rail, with a journey duration of less than approximately three hours (e.g., France, Belgium, Austria). Since ultra-short-haul flights generate twice as many emissions per seat-kilometre than long-haul flights, and they can be replaced with less polluting high-speed rail (HSR) options, it can be said that such an initiative is justified, Grimme and Jung (2018). Therefore, it is not surprising that many national authorities at the national and EU levels started to consider the idea of banning ultra-short-haul flights and replacing them with HSR options (e.g., Netherlands, Spain and Germany).

As an alternative solution to preserve their position on ultra-short-haul market, and to return air transport to banned routes, airlines could consider the introduction of new electrical vehicles. These vehicles, Electric Vertical Takeoff and Landing Aircraft (eVTOLs), characterized by lower pollution, will be able to cover certain origin and destination pairs and potentially enable the feasibility of the previously banned routes. The majority of research papers consider on-demand urban air mobility, but not the introduction of eVTOLs on ultra-short-haul routes. Some papers also provide a review of the current technology and research emphasizing the advantages and challenges of eVTOLs, Vieira et al. (2019), Polaczyk et al. (2019), Barzkar and Ghassemi (2020).

It is worth noting that the catchment area for an eVTOL aircraft depends on its range. Therefore, it is of paramount importance for the airline to adequately select the aircraft, as it will directly affect the competitive positions and sustainability of airlines. However, aircraft selection is a complex problem of strategic importance even in the case when needed data are available. Bearing in mind that the vast majority of eVTOLs are in the prototype testing or developing phase, the data availability is questionable, and that brings additional complexity. The costs, as one of the very important criteria are hard to assess, and accordingly, cannot be considered in the process of fleet selection. To make a satisfactory choice while dealing with multiple criteria, the Analytic Hierarchy Process (AHP) approach can be employed to support eVTOL selection. Since data on costs are not available, the AHP can help conduct sensitivity analysis to assess the influence of costs on aircraft ranking and selection.

The main objective of this paper is to reveal how costs influence the eVTOL aircraft selection. More precisely, the sensitivity of eVTOLs rankings concerning different pairwise comparisons of the alternatives with respect to costs is analysed, showing the influence of cost changes on aircraft selection.

2. Contextual settings

2.1. Regulatory framework towards net-zero carbon aviation

In an effort to effectively combat the climate change and reduce the region's carbon footprint, certain countries have begun encouraging their national airlines to transition from air travel to ground transportation by offering government-issued vouchers. Meanwhile, others have taken a more restrictive approach by introducing bans on short-distance flights. Among different countries which considered this measure as a viable option, France was a leading one enacting a ban on short-haul flights on routes where the train is feasible alternative with a travel duration of two and a half hours or less. Under the Climate and Resilience law of 22 August 2021, the country set out to reduce greenhouse gas emissions by 40% over the current decade. Article 145, Section II of the law specifies that “*scheduled public passenger air transport services are prohibited on all air routes within French territory whose route is also covered on the national rail network without connections and several daily services with a duration of less than two hours and thirty minutes*”, Légifrance (2021). For a flight to fall under this ban, specific conditions must be met:

- The train journey must be direct, with no station transfers allowed;
- Flights that are part of a longer, connecting itinerary are exempt;
- The rail service must offer appropriate frequency and scheduling, enabling travellers to spend at least eight hours at their destination during the day, with trains available in both the early morning and late evening.

Prior to the 2021 legislation, the French government had already required its national airline, Air France, to reduce certain short domestic routes as a condition for receiving a €7 billion (\$7.9 billion) state aid package during the

COVID-19 pandemic in 2020. The subsequent law introduced in 2021 broadened this requirement, extending the restrictions to all carriers operating within French territory, Condé Nast Traveler (2022).

Based on the current capabilities of the national rail infrastructure and the criteria outlined in the decree, only a limited number of flight routes have been officially replaced with rail alternatives. These include: Paris-Orly (ORY) to Bordeaux (BOD), Paris-Orly (ORY) to Nantes (NTE) and Paris-Orly (ORY) to Lyon (LYS).

Each of these connections is now serviced by France's high-speed TGV rail network. Passengers traveling to Bordeaux and Nantes depart from Montparnasse station in Paris, while those heading to Lyon begin their journey at a different station. Because Paris-Orly Airport is located relatively close to the city center, transfers to these train stations are short, resulting in total journey times—station transfer plus train ride—of less than 150 minutes. In addition, the track from central Paris to Lyon is not entirely high-speed from the beginning, but since trains reach speeds that exceed 200 km/h further along the route, the trip is able to be completed within the two-and-a-half-hour threshold.

As of 2024, France remains the only EU member state to enforce a nationwide policy explicitly designed to promote a shift from air to rail for environmental reasons. Nonetheless, other countries such as Germany, Austria, Belgium, the Netherlands, and Spain are also introducing legislative measures aimed at encouraging more sustainable travel. Spain's *España 2050 Strategy* explicitly recommends adopting measures similar to those in France to address the climate concerns. In contrast, Central European countries like Austria have not imposed direct bans based on rail alternatives. However, in 2020, the Austrian government provided a €450 million state-backed bond to Austrian Airlines as part of a post-COVID-19 support package, Transport and Environment (2020). In exchange, the airline agreed to replace short-haul flights with train routes where the rail journey takes under three hours—including those involving connecting flights—going beyond France's more limited criteria of 2.5 hours and direct rail only.

However, it is worth mentioning that flights with a radius of less than 500 km accounted for 30% of the total European traffic. In addition, some of these flights do not have a viable train alternative. Thus, the introduction of new electrical aircraft on these routes is seen as an effective way in achieving less carbon emission transportation.

2.2. Fleet selection problem

Among various multi-criteria decision-making (MCDM) methods used for aircraft selection, the Analytic Hierarchy Process (AHP) remains prominent due to its simplicity and effectiveness. Notable studies include Bruno et al. (2015), who integrated AHP with Fuzzy Sets, and Dožić and Kalić (2015), who explored aircraft selection and fleet planning through AHP, the even swap method, and fuzzy approaches. Ozdemir and Basligil (2016) introduced advanced techniques like the fuzzy Analytic Network Process (ANP) and Choquet integral to capture complex interdependencies among criteria. More recently, Dožić et al. (2018) applied the logarithmic fuzzy preference programming (LFPP) method for regional aircraft evaluation. Across these studies, selection criteria generally fall into four categories: technological, economic, service quality, and environmental factors. While economic data can be difficult to access due to confidentiality, the growing emphasis on sustainability has elevated environmental criteria in recent fleet planning.

In light of the emergence of new vehicle types, the problem of fleet selection is expected to become even more prominent, as these aircraft must be integrated into existing airline fleets. The operational costs of these new aircraft remain uncertain, as they will depend not only on technical factors but also on the business strategies adopted by future operators. However, the current methodology provides a solid foundation that can help managers consider various indicators and metrics relevant to these future operations.

3. Methodology

The methodology applied in this paper consists of four steps, Fig. 1: 1) definition of sets of criteria/sub-criteria and set of alternatives, 2) pairwise comparison matrices definition, 3) selection of an eVTOL for ultra-short-haul route applying AHP, 4) sensitivity analysis and results discussion.

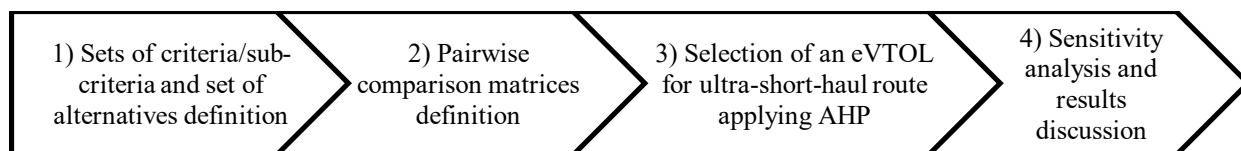


Fig. 1. Methodology.

1) To define sets of criteria/sub-criteria the previous research, Dožić (2019) is used and data availability is considered. Initially, four criteria were selected: seat capacity, maximum take-off mass (MTOM), maximum payload capacity and range of eVTOL. Seat capacity reflects the capability of keeping a market share of an airline. Greater capability is closely connected to greater profitability. MTOM is the main unit used to calculate the airport and navigation fees (measured in kilograms). Maximum payload capacity can be considered as a measure of level of service from the passengers' perspective. The higher payload capacity is offered, the better is. Range defines the eVTOL's ability to fly a certain distance without recharging the battery (measured in kilometers). Depending on the eVTOL range, the set of ultra-short-haul routes that can be served is determined. After conducting the short survey, respondents who are experts in different fields of aviation pointed out that besides the criteria mentioned, costs should be taken into consideration, as well, and finally, five criteria, including costs, are defined. In this research, criterion costs include all costs related to an eVTOL. Although the data on costs are unavailable, this criterion will contribute to the analysis showing the impact of cost changes on the final ranking of eVTOLs. Concerning the set of alternatives, the main constraint is the availability of data, since eVTOLs are in the development or prototype-testing phase. Additionally, the goal was to select an eVTOL that will operate ultra-short-haul routes that are banned in some European countries, to help airlines keep their market position. In accordance with that, the required capacity is set to be greater than two passenger seats. The search for eVTOLs and the available data on their characteristics resulted in the set consisting of six eVTOLs: Joby S4, Lilium Jet, Terafugia TF-2 Lift+Push, Alice, Jaunt, and Embraer Flexcraft.

2) Pairwise comparison matrices are created using the experts' opinions obtained from the previously mentioned survey, but only for the initial four criteria. Since the data related to costs are unavailable, the costs are set to have the same (the highest) importance as capacity and MTOM in the basic solution, in the first level of hierarchy, while in the second level of hierarchy, all eVTOLs have the same costs in the basic solution.

3) The AHP is one of the most commonly used MCDM methods in aviation, Dožić (2019), therefore, it is employed to find out which eVTOL is the most appropriate solution for preserving ultra-short-haul routes. The best ranking eVTOL, according to the basic solution is Alice.

4) To find out how costs influence eVTOL selection, the sensitivity analysis is performed. As it is mentioned, there is a lack of data related to any costs (purchasing costs, unit costs, etc.). The AHP approach provides a pairwise comparison of alternatives with respect to criteria, and this feature helps assess the cost influence on the selection of eVTOL, even if there is no data. In the basic solution, all costs are assumed to be equal (since data are unavailable), but sensitivity analysis gives a deep insight into their influence. In sensitivity analysis, all six aircraft are assumed to have different costs. The number of experiments performed corresponds to the number of permutations without repetition of aircraft order with respect to costs, which is equal to $6! = 720$, representing the set of all feasible solutions.

4. Sensitivity analysis and results discussion

This section is dedicated to the sensitivity analysis of a basic solution derived concerning cost changes. Before presenting the analysis and providing a discussion, the selection of an eVTOL (e.g., a basic solution) applying the AHP should be given.

4.1. Basic solution

The values for selected criteria (except the costs), for all six eVTOLs, are given in Table 1. The data are collected from the manufacturers' websites as well as from other relevant sources. The data given in Table 1 are used to create pairwise comparison matrices of eVTOLs with respect to different criteria.

Table 1. The values for selected criteria.

	Capacity	MTOM (t)	Maximum payload (kg)	Range (km)
Joby S4	4	2404	453	161
Lilium Jet	6	3175	500	250
Terafugia TF-2 Lift+Push	4	4080	500	400
Alice	9	7485	1179	402
Jaunt	4	1814	362	182
Embraer Flexcraft	7	3239	1000	926

A pairwise comparison matrix of selected criteria, Table 2, is created based on experts' opinions (12 experts employed in different sectors in the aviation industry, e.g., airlines, ANSP, regulatory bodies, drone pilots, and universities who rated their expertise on average by 3.3 out of 5, Dožić et al. (2026)). The priorities for each criterion are given in Table 2. Since the matrix is consistent (consistency ratio $CR < 0.1$), the local priorities can be used to calculate the global priorities for eVTOLs in the basic solution. Table 3 shows local priorities for the set of alternatives with respect to all criteria selected, as well as the global priorities aggregated from the local priorities (last column).

Table 2. Pairwise comparison matrix of selected criteria.

	Capacity	MTOM	Maximum payload	Range	Cost	Local priorities
Capacity	1	1	3	3	1	0.2734
MTOM	1	1	1	3	1	0.2254
Maximum payload	1/3	1	1	3	1/3	0.1527
Range	1/3	1/3	1/3	1	1/3	0.0751
Cost	1	1	3	3	1	0.2734
$\lambda_{max}=5.2306$		$CI=0.0577$		$CR=0.0515$		

Table 3. Aggregation of the local priorities.

	Capacity	MTOM	Maximum payload	Range	Cost	Global priorities
Joby S4	0.0813	0.2304	0.1090	0.0574	0.1667	0.1407
Lilium Jet	0.1456	0.1364	0.1090	0.0940	0.1667	0.1398
Terafugia TF-2 Lift+Push	0.0813	0.0812	0.1090	0.2135	0.1667	0.1188
Alice	0.3719	0.0519	0.3049	0.2135	0.1667	0.2215
Jaunt	0.0813	0.3637	0.0631	0.0630	0.1667	0.1641
Embraer Flexcraft	0.2387	0.1364	0.3049	0.3586	0.1667	0.2151

According to values of RIs , all created pairwise comparison matrices of eVTOLs with respect to different criteria are consistent. The final rank of six eVTOLs and their global priorities, in the basic solution, are as follows: Alice (0.2215), Embraer Flexcraft (0.2151), Jaunt (0.1641), Joby S4 (0.1407), Lilium Jet (0.1398) and Terafugia TF-2 Lift+Push (0.1188).

4.2. Sensitivity analysis

After reaching the basic solution (where all aircraft are assumed to have the same costs), the sensitivity of the solution (rank of alternatives) concerning different costs in the comparison matrix of alternatives with respect to costs is to be analyzed. Assuming that all eVTOLs have different values of costs and setting the lowest cost for an eVTOL aircraft, it is possible to derive 120 different experiments for each eVTOL (in total 720 experiments, representing the set of all feasible solutions). This provides the complete picture of cost influence on the rank of alternatives and the most appropriate solution, as well. Pairwise comparison matrices of eVTOLs with respect to costs, for all experiments,

are generated by creating the Python code. This code is also used to determine the appropriate rank of six eVTOL aircraft (to calculate all global priorities) and to check the consistency of matrices (all *CR* values are calculated, and they are satisfying, below 0.1).

Fig. 2 presents the basic solution (experiment 0) and 120 experiments in the case when Joby S4 is the eVTOL aircraft with the lowest costs (experiments 1-120). The experiment 0 presents the case where, in the absence of information, is assumed that all costs are equal. The sensitivity analysis assumed that all aircraft have different costs. In experiment 1, the assumption is that Joby S4 has the lowest cost (1), and the order of the other eVTOLs based on the costs is Lilium Jet (2), Terafugia TF-2 Lift+Push (3), Alice (4), Jaunt (5), and Embraer Flexcraft (6) (the highest costs). The permutation that corresponds to experiment 1 is 123456. For experiment 2, the corresponding permutation is 123465, which means that it is assumed that the last two aircraft switch their positions according to costs. The next permutation is 123546, and all other experiments and corresponding permutations are sorted in increasing order, concluding with the last one, which is 165432. The best-ranked aircraft are Alice (in 54 experiments), Embraer Flexcraft (in 42 experiments), and Joby S4 (in 24 experiments). The eVTOLs ranked in the sixth position are Terafugia TF-2 Lift+Push (in 84 experiments), Lilium Jet (in 32 experiments) and Jaunt (in 4 experiments). The highest value of importance is equal to 0.2440 and it belongs to Alice, while the lowest value of importance is equal to 0.0851 and is assigned to Terafugia TF-2 Lift+Push.

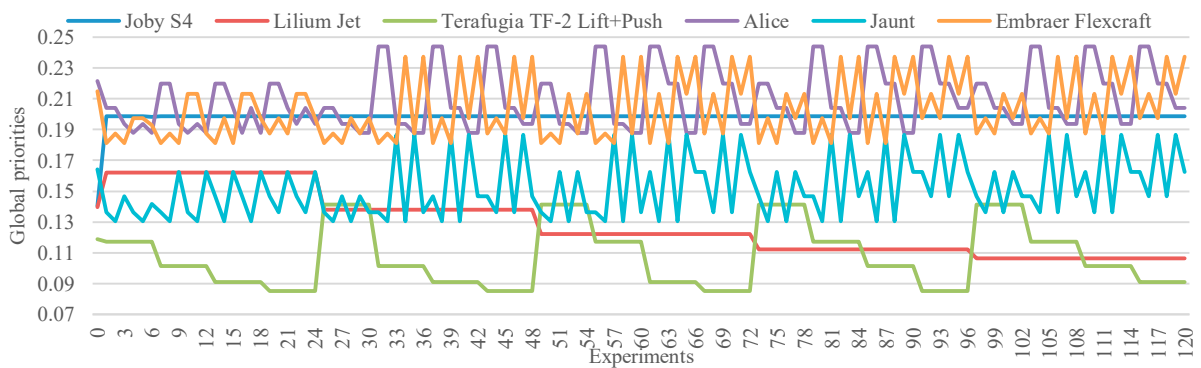


Fig. 2. eVTOLs aircraft rankings with respect to order of costs changes (Joby S4 – the lowest costs).

Fig. 3 presents the case when the bottom-ranked eVTOL aircraft in the basic solution (Terafugia TF-2 Lift+Push) is set to have the lowest cost. Experiment 1 corresponds to permutation 231456 (the order of aircraft based on costs, starting from the one with the lowest costs, is: Terafugia TF-2 Lift+Push, Joby S4, Lilium Jet, Alice, Jaunt, Embraer Flexcraft).

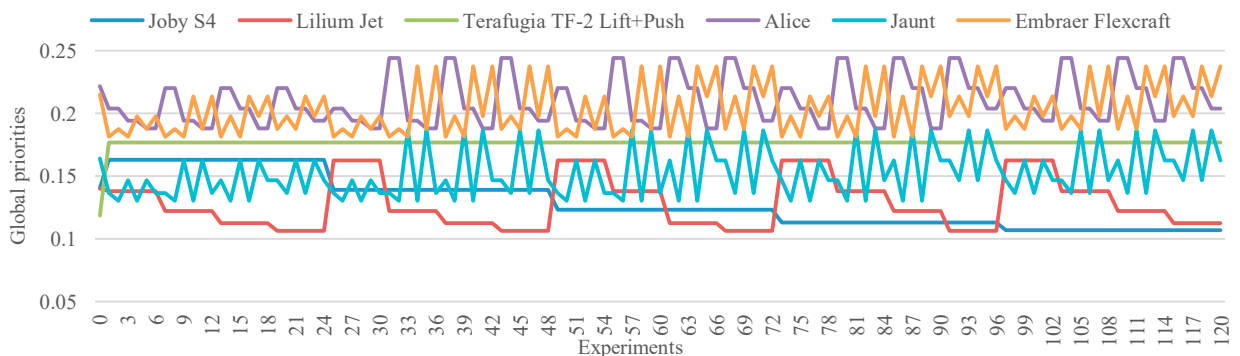


Fig. 3. eVTOLs aircraft rankings with respect to order of costs changes (Terafugia TF-2 Lift+Push – the lowest costs).

All permutations have 1 at the third position, and again, they are sorted in increasing order, starting from 231456 to 651432. It is shown that despite the lowest cost, this aircraft can never be the most appropriate solution, and that is the result of the defined pairwise comparison matrix of selected criteria. The best position reached by Terafugia TF-2 Lift+Push is the third (in 96 experiments), while it is the fourth-ranked in 24 experiments. The top-ranked are Alice (in 66 experiments) and Embraer Flexcraft (in 54 experiments). More precisely, these two are switching on the first and the second positions. The bottom-ranked are Joby S4 (in 56 experiments), Lilium Jet (in 54 experiments) and Jaunt (in 8 experiments).

Fig. 4 demonstrates the dominance of the most appropriate eVTOL aircraft in the basic solution (Alice) over all other eVTOLs in the case when it has the lowest cost assigned. Experiment 1 corresponds to permutation 234156 (the order of aircraft based on costs, starting from the one with the lowest costs, is: Alice, Joby S4, Lilium Jet, Terafugia TF-2 Lift+Push, Jaunt, Embraer Flexcraft). All permutations have 1 at the fourth position (since Alice has the lowest costs), and again, they are sorted in increasing order, starting from 234156 to 654132. It is also shown that the bottom-ranked aircraft are Terafugia TF-2 Lift+Push (in 76 experiments), Lilium Jet and Joby S4 (each of them in 22 experiments).

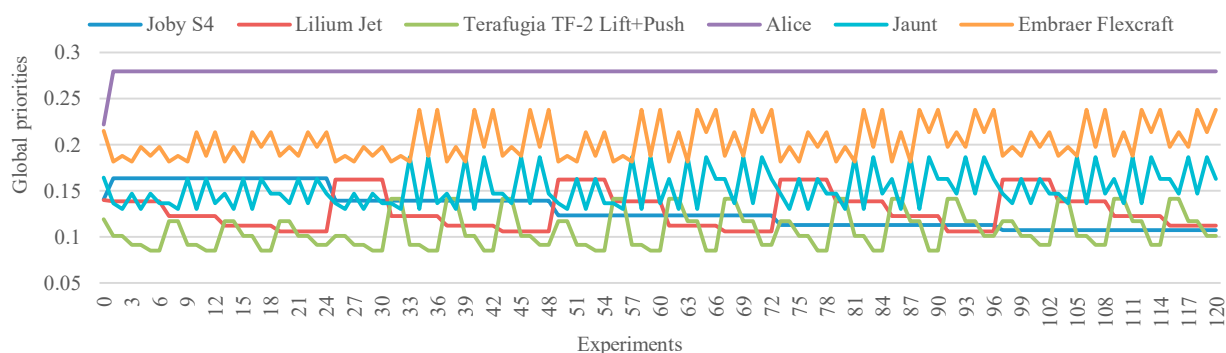


Fig. 4. eVTOLs aircraft rankings with respect to order of costs changes (Alice – the lowest costs).

It can be seen that if the eVTOL Joby S4 has the lowest costs assigned (it is the fourth ranked in the basic solution), it changes its position depending on the costs of other aircraft and it can be the best, but not the worst solution (Fig. 2), that the eVTOL Terafugia TF-2 Lift+Push which is ranked as the last one in the basic solution is never the best solution even if it has the lowest costs (Fig. 3), and that if the eVTOL aircraft which is ranked as the best solution in the basic solution will always be the best solution if it has the lowest costs (Fig. 4).

In the case when the lowest cost is assigned to Embraer Flexcraft, this aircraft is observed as dominant over all other alternatives, while in the second position is always Alice. The bottom-ranked eVTOLs are Terafugia TF-2 Lift+Push (in 76 experiments), Lilium Jet and Joby S4 (each of them in 22 experiments). If the lowest cost is assigned to Lilium Jet, three eVTOL aircraft are the top-ranked, and they are Alice (in 54 experiments), Embraer Flexcraft (in 42 experiments) and Lilium Jet (in 24 experiments), while other three aircraft Terafugia TF-2 Lift+Push (in 84 experiments), Joby S4 (in 32 experiments) and Jaunt (in 4 experiments) are bottom ranked. It should be noted that the position of Lilium Jet can be the first (in 24 experiments), second (in 72 experiments) and third (in 24 experiments). Finally, if the lowest cost is assigned to Jaunt, then Jaunt is most commonly the top-ranked (in 72 experiments), followed by Alice and Embraer Flexcraft (each in 24 experiments). The bottom-ranked are Terafugia TF-2 Lift+Push (in 76 experiments), and Joby S4 and Lilium Jet (each in 22 experiments). Despite the lowest cost, Jaunt can be ranked in the second position (in 48 experiments).

Considering all 720 experiments, it can be observed that Alice (318) and Embraer Flexcraft (282) are most commonly ranked as the best solution, and they are followed by Jaunt (72), Joby S4 (24) and Lilium Jet (24). On the other hand, Terafugia TF-2 Lift+Push is ranked 396 times as the worst solution, and it is followed by Lilium Jet and Joby S4, which are bottom-ranked in 154 experiments (each of them), while Jaunt appears as the worst only in 16 experiments. The highest value of importance in all experiments is equal to 0.2797 and it is assigned to Alice, while the lowest value of importance is equal to 0.0851 and is assigned to Terafugia TF-2 Lift+Push. The results show that Alice and Embraer Flexcraft are two eVTOL aircraft that are never ranked as the worst solution, while Terafugia TF-

2 Lift+Push never reaches the best position. Joby S4, Lilium Jet and Jaunt are the most influenced by costs, and depending on the costs, these three eVTOL aircraft could be ranked in the best and the worst positions as well.

5. Conclusion and future work

Although the AHP approach could be considered a basic approach, it is a very powerful tool in the case of incomplete data, and it can manage it in a very successful manner, giving a practical contribution to the problem of eVTOL selection. The sensitivity analysis of eVTOLs ranking concerning the costs shows that the final rank of six aircraft is sensitive to cost changes. Moreover, it is revealed that Terafugia TF-2 Lift+Push, can never be chosen as the most desirable solution for the defined pairwise comparison matrix of criteria selected (the first level of hierarchy), Alice and Embraer Flexcraft can never be bottom-ranked, while the other three aircraft can be both the best and the worst solution.

Future work could extend the sensitivity analysis by changing the pairwise comparison matrix for the first level of the hierarchy, i.e., changing the importance of criteria over each other. Additionally, deeper analysis could be provided by assuming that costs could be equal for certain aircraft, which will significantly increase the number of experiments (permutations with repetition). Although some European countries are considering extending rail alternatives to replace flights under five hours, data from various national transport agencies indicate that the majority of aviation-related emissions across Europe stem from international rather than domestic flights. Consequently, shifting short-haul routes to rail is unlikely to produce substantial emission reductions on its own. It implies that more comprehensive strategies addressing medium- and long-haul air travel will be necessary to achieve meaningful climate goals. The given methodology and sensitivity analysis can allow for more informed aircraft selection on these longer routes.

Acknowledgements

This research has been supported by SESAR Joint Undertaking and European Commission, and has been done within the projects “Implemented Synergies, data sharing contracts and Goals between transport modes and AIR transportation” (SIGN-AIR) under grant agreement No. 101114845. Part of this study has been supported by the Ministry of Science, Technological Development and Innovation, Republic of Serbia.

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